Environmental Predictors of Reported Crop Cultivation and Yield in Gujarat: A Five-Crop District-Level Study

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***Abstract*—Understanding where crops are cultivated is distinct from explaining how much they yield. We examine district-level agricultural statistics for castor, cumin, fennel, fenugreek, and isabgol across 33 districts of Gujarat, combining government area and production records with NASA POWER weather and gridded soil estimates. A consolidated panel contains 1,031 district–crop–year observations over eight agricultural years; 682 and 540 observations qualify for preferred-source and strict yield analyses, respectively. Under five geographic-block folds, a crop-and-environment logistic model attains an area under the receiver operating characteristic curve (AUC) of 0.906, versus 0.744 for a crop-only baseline. A district-resampled interval for the AUC difference is 0.081–0.240. By contrast, ridge regression, random forests, and XGBoost do not outperform a crop–year mean baseline for log-yield prediction. Bayesian hierarchical regressions converge adequately but pooled environmental coefficients have intervals spanning zero. For district cultivation-rate interpolation, spatial-block ordinary Kriging yields root mean squared error (RMSE) 0.204, compared with 0.255 for inverse-distance weighting and 0.311 for the training mean. These results distinguish predictive cultivation patterns from weaker evidence about environmental drivers of yield. The findings concern reported district-level cultivation, not field-scale suitability or causal effects.**

***Index Terms*—agricultural statistics, Gujarat, cultivation occurrence, Bayesian hierarchical model, spatial cross-validation, Kriging, interpretable machine learning**

I. Introduction

Crop distribution and crop productivity are related, but they answer different questions. Cultivation decisions reflect agroecological conditions alongside markets, irrigation, infrastructure, and management. Yield responds to additional factors that may not be captured by district-average climate and soil maps. A model that predicts the presence of cultivation therefore should not automatically be interpreted as a mechanistic model of productivity or intrinsic suitability.

This study evaluates five economically and medicinally relevant crops in Gujarat: castor, cumin, fennel, fenugreek, and isabgol. The principal questions are: (i) whether crop and environmental descriptors predict *reported cultivation occurrence* in geographically held-out districts; (ii) whether those descriptors improve prediction of reported yields beyond crop–year benchmarks; and (iii) whether spatial interpolation of district cultivation rates remains useful under geographic-block validation. We also quantify the sensitivity of Bayesian environmental associations to competing government source selections.

The contribution is a transparent comparison of predictive tasks using the same district-level research panel, with explicit baselines, source-linked zero-label checks, and geographic validation. The analysis does not claim causal identification, agronomic threshold confirmation, or farm-level mapping.

A. Research context and questions

The original course study was motivated by medicinal-plant site selection under biotic and abiotic stress. Its initial shortlist included ashwagandha, tulsi, aloe vera and senna; those species were not retained because the project could not document compatible district-level yield series. The final five-crop selection instead balances medicinal or aromatic relevance with the availability of administrative crop statistics. This decision limits botanical breadth but allows the proposed predictions to be checked against recorded outcomes rather than against synthetic recommendation labels.

An important distinction emerged during implementation. An administrative report records a district's cultivated area and production; it does not measure whether uncultivated land would support a species if planted. The outcome in this paper is therefore *reported cultivation*, not an intrinsic ecological niche or a recommendation to a particular farmer. Similarly, government yield statistics are averages over farms and management regimes rather than controlled experiments. The central methodological question is whether these observational signals can be separated with adequate validation.

The four stages were designed to answer increasingly demanding questions: construction of a reproducible source-linked panel; hierarchical estimation of associations between environmental variables and yield; predictive comparison with explainable machine learning; and spatial grouping and interpolation. Later validation audits add source-cell checking and geographic holdouts. These additions are substantive changes from the first course report, not merely revised figure formatting.

B. Relationship to prior work

Roopashree *et al.* examined soil-based medicinal-plant matching in Karnataka using geographic and ensemble-learning methods [9]. Their setting motivates the regional comparison but is not an experimental baseline for the present five-crop Gujarat panel. Crop recommendation studies such as Dey *et al.* combine nutrients and climatic covariates, but predictive

accuracy on a curated recommendation dataset cannot be equated with independent validation on historical government APY records [10]. Jeong *et al.* demonstrated the potential of random forests for regional crop-yield modeling while highlighting the dependence of skill on the study setting [11]. Here the explicit distinction between presence and yield, rather than a new classification algorithm, is the central analytical contribution.

Spatial validation is crucial because neighboring districts share agroclimatic conditions and sometimes share the same gridded weather estimate. Roberts *et al.* discuss the limitations of conventional cross-validation in spatially and temporally structured data [12]. Our held-out-district and held-out-region experiments address part of this issue, while remaining internal validations within Gujarat. The analysis also uses SHAP as a descriptive explanation of fitted models [6], not as a causal attribution method. Hierarchical Bayesian estimation is used to represent uncertainty and partial pooling rather than to assert that every crop responds to a common environmental threshold [8], [13].

II. Data and Evidence

A. Government crop statistics

The analysis uses Gujarat Directorate of Agriculture area, production, and yield (APY) reports and the Directorate of Arecanut and Spices Development's *Spices Statistics at a Glance 2021* [1], [2]. District names were harmonized across publications. Government evidence was retained separately from the canonical modeling panel so that conflicting observations remain auditable. The panel has 1,031 district–crop–year keys across 33 districts, five crops, and eight agricultural years. Its cultivation labels comprise 696 positive-area observations and 335 reported zero-area observations.

A source-cell audit supplied with the project reports agreement for all 335 zero-area labels with the relevant original APY workbook or PDF entries. This establishes consistency with published zero entries, not proof of physical absence of cultivation. Nonzero yield records have not all been independently re-transcribed; competing government yield estimates remain an uncertainty source. The preferred-source yield subset contains 682 observations, and the stricter conflict-excluded subset 540.

B. Weather, soil, and geographic scale

Daily NASA POWER point estimates were aggregated over crop-specific candidate cultivation windows [3]. Temperature, rainfall, humidity, and derived weather indicators were linked to district–crop–year observations. District soil descriptors include pH, texture fractions, and soil organic carbon from gridded products. These are modeled environmental estimates rather than field samples. A corrected polygon-interior coordinate was used for Devbhumi Dwarka's weather re-extraction. Spatial predictors do not represent the full within-district heterogeneity of agricultural land.

TABLE I
Research panel and audit populations

Quantity	Count
Districts / crops / agricultural years	33 / 5 / 8
Canonical district–crop–year observations	1,031
Reported cultivated / zero-area labels	696 / 335
Preferred / strict yield observations	682 / 540
Spatial validation blocks	5

C. Evolution from the original course datasets

The initial project maintained four separate data products: a district–plant master table, a documented fenugreek yield table, a monthly Gujarat weather series, and a four-plant yield–climate panel. These files were useful for prototyping but mixed temporal and spatial resolutions. The first course methodology described 165 district–crop profiles and a 426-row analytical panel: three agricultural years for four crops and a single documented fenugreek year. In that version, weather was aggregated over a common July–June period, soil properties were often categorical or inferred from published thematic maps, and newer districts sometimes inherited parent-district classes. These choices were stated explicitly in the submitted course methodology and should not be confused with the later research release.

Subsequent dataset revisions prioritized verifiable source observations. A candidate fenugreek series was rejected after the project audit found repeated district values derived from statewide totals rather than separately reported observations. The V6 release retained source-level evidence and assembled four new data products: (1) a five-crop district registry with district soil descriptors; (2) government yield source evidence; (3) corrected-coordinate NASA daily weather; and (4) an integrated district–crop–agricultural-year research panel. Thus the four V6 files are successors to, not row-for-row copies of, the original four. Earlier findings about a three-year panel, coarse pH classes, seven HDBSCAN groups, or a large cumin heat coefficient are *historical* results and are not substituted into the current V6–V9 findings.

D. Source acquisition and record lineage

Gujarat's Directorate of Agriculture publishes district-wise APY reports in both spreadsheet and PDF formats. The archived workbooks have original download-style filenames; the consolidated evidence table records the source filename, sheet or page, row/column location, agricultural year and a source reference for each observation. The 2015–2018 final-forecast PDF is especially valuable because it contains district rows for castor, cumin, fennel and isabgol, and identifies its area and production units as hundreds of hectares and hundreds of metric tonnes, respectively. The published yield column is expressed in kg/ha. Source-specific unit conversion is performed before joining; no common conversion factor is applied to all publications without checking their headings.

For cumin, fennel and fenugreek, the Directorate of Arecanut and Spices Development volume supplies an additional published series. The data integration retains overlapping

TABLE II
EVOLUTION OF THE RESEARCH DATA PRODUCTS AND SAFEGUARDS

Component	Original course implementation	Current V6 research implementation
Unit of observation	165 district–crop profiles; 426 available district–crop–year rows	1,031 unique district–crop–year observations; explicit source and eligibility fields
Crop yield	Three years for four crops; one documented fenugreek year	Up to eight agricultural years depending on crop; preferred and strict government-source subsets
Climate	District-point monthly NASA POWER; July–June summary	Daily NASA POWER linked to candidate crop windows; heat-day, rainfall and humidity features
Soil	Dominant categorical map classes and pH midpoint approximations	District-polygon aggregated gridded soil attributes, including continuous pH, texture fractions and modeled carbon
Quality controls	Preliminary duplicate and plausibility screening	Source precedence, conflict flags, duplicate-key resolution, coordinate review and separate zero-label verification
Evaluation	Primarily leave-one-district-out	Leave-one-district-out plus five geographic blocks and district-bootstrap intervals

source rows rather than averaging disagreements. Material conflicts are flagged, a documented source precedence determines the preferred value, and the strict subset excludes records whose evidence does not meet the stronger eligibility rule. The exact precedence policy is an empirical data-engineering decision, not evidence that one government estimate is infallible. A fully reproducible release should archive the versioned source-selection table alongside the paper.

A critical safeguard is distinguishing a numeric zero from a blank cell. Area equal to zero, with production also reported as zero, defines a reported absence label in the panel. Missing source cells do not automatically define noncultivation. The project's final source-cell audit checked 335 zero labels against specified original spreadsheet cells or PDF district rows and reported 335 matches, with nine district-name normalizations documented during checking. This validates the transcription of the reported zeros, but it cannot establish absence at individual fields or the completeness of the administrative reporting process. Independently checking a stratified sample of positive-area and yield values remains important for journal submission.

E. District harmonization and boundary policy

Names differ between sources: examples include Kachchh/Kutch, Ahmadabad/Ahmedabad, Mehesana/Mehsana, Arvalli/Aravalli and Chhotaudepur/Chhota Udaipur. The integration therefore uses a canonical 33-district name list and preserves original source names in separate columns. District boundaries were obtained as administrative polygons and checked against a name crosswalk. A Devbhumi Dwarka coordinate used for weather extraction was corrected and re-extracted. The Stage 1 audit still reports 32 coordinate-review flags; the exact meaning and disposition of those flags should be preserved in the release rather than described as universally resolved.

A present-day 33-district map is not identical to every historical administrative reporting geography. Consequently, joining historical district statistics to contemporary polygons requires a documented harmonization policy. The study is explicitly district-scale; it does not infer that a point soil estimate or district-mean weather observation represents every cultivated field.

F. Climate feature construction

The daily NASA POWER archive provides point-based estimates for each district sampling coordinate [3]. The project builds a daily series before aggregating to crop–year windows. For a candidate sowing date s_{ct} and harvesting date h_{ct}, accumulated precipitation and mean temperature are calculated as

$$P_{dct} = \sum_{\tau=s_{ct}}^{h_{ct}} p_{d\tau}, \tag{1}$$

$$\overline{T}_{dct} = \frac{1}{N_{dct}} \sum_{\tau=s_{ct}}^{h_{ct}} T_{d\tau}, \tag{2}$$

$$H_{dct}(q) = \sum_{\tau=s_{ct}}^{h_{ct}} \mathbb{1}(T_{\max,d\tau} > q), \tag{3}$$

where N_{dct} is the count of observed days. Relative humidity and solar-radiation summaries are also retained, together with alternative calendar windows and completeness flags. The exported dataset records thresholds for 30, 35 and 40 °C rather than asserting that each threshold is a crop-specific biological injury point. Candidate calendars are assumptions, not observed sowing dates. The use of a single weather point per district can produce repeated covariates in adjacent districts and reduces effective spatial information.

G. Gridded soil and raster coverage

The revised environmental dataset uses polygon summaries of available gridded soil rasters rather than treating a district-center sample as a district mean. Soil pH, sand, clay and organic-carbon descriptors are accompanied by raster valid-pixel counts and coverage fractions. The original Earth Engine source-mask check was needed because successful API responses alone do not establish complete raster coverage. OpenLandMap's organic-carbon product describes its stored values using a factor of five grams per kilogram; scaling must

TABLE III
FOUR INTEGRATED V6 RESEARCH FILES

Dataset	Role in analysis
District registry and soil	Canonical crop–district combinations, polygon identifiers, soil attributes
Government source evidence	Source-row APY values, provenance and conflict tracking
Daily corrected weather	District point weather series and coordinate metadata
District–crop–year panel	Model-ready joins, source eligibility, seasonal weather and fold definitions

be applied before interpreting its units [4]. The final dataset contains modeled estimates at specified depths, not laboratory analyses of sampled farms. Soil is largely time-invariant in this panel, so soil coefficients can also reflect persistent district differences.

III. METHODS

A. Cultivation occurrence and held-out geography

For each district d, crop c, and agricultural year t, the binary response is $z_{dct} = \mathbb{1}(A_{dct} > 0)$ when an APY area entry is available. The primary classifier is regularized logistic regression with crop identity and environmental covariates. The crop-only classifier is the benchmark. All imputation and scaling are fitted within training folds. The geographic-block experiment holds out each of five preassigned district blocks; no district is shared between training and test partitions. The blocks are provisional and the results are not independent external validation. We report pooled out-of-fold AUC, average precision (AP), and Brier score. District-level bootstrap resampling (500 replicates) provides conditional uncertainty for the AUC difference.

B. Yield prediction and Bayesian association

Positive yield observations are analyzed on the logarithmic scale. Ridge, random forest, and XGBoost models are compared with a crop–year mean baseline under district-held-out prediction. Pooled R^2 can be dominated by differences among crops and years; therefore, superiority to the crop–year baseline is the relevant incremental-prediction criterion.

A hierarchical Bayesian Student-t model uses crop intercepts, partially pooled crop-specific slopes for standardized temperature, rainfall, and pH, district random intercepts, and agricultural-year effects:

$$\log Y_{dct} \sim t_{\nu=5}(\mu_{dct}, \sigma), \quad \mu_{dct} = \alpha_c + \boldsymbol{\beta}_c^{\top} \mathbf{x}_{dct} + u_d + v_t. \tag{4}$$

Separate models were fitted to preferred and strict subsets with four Markov chains, 1,000 tuning iterations and 750 retained draws per chain. These Bayesian summaries are associational and were not used as held-out yield predictions.

C. Environmental zoning and interpolation

District-averaged weather and soil features were standardized and clustered into three exploratory environmental zones using Ward agglomerative clustering. Associations with district cultivation rates were evaluated descriptively with η^2 and label permutations. For spatial interpolation, ordinary Kriging with a spherical variogram, inverse-distance weighting (IDW), and the training-block mean were evaluated on the same five geographic holdouts. Kriging variogram fitting uses training districts only. The response is district cultivation rate; interpolated surfaces are not field-level suitability estimates.

D. Feature preparation and missing-data rules

Predictors are grouped into crop identity, seasonal weather and district soil descriptors. Presence is defined from the observed area field, while yield is modeled only when a positive, eligible source observation exists. This prevents the false interpretation of uncultivated districts as farms experiencing zero productivity. Data leakage is addressed by fitting any imputation, scaling or encoding steps inside the training portion of each validation fold. The project does not include geographical coordinates as direct cultivation-model predictors merely to memorize district locations; geographic information instead determines which districts are held out together.

An earlier course implementation constructed a range-gap stress index, $S(v; L, U) = \max(L - v, 0) + \max(v - U, 0)$, from published crop requirements. The current research panel retains more directly observed seasonal climate variables and their transformations. Unless crop-specific optimum ranges are independently documented and the current code actually computes those indices, the present Bayesian slopes should be described as associations with model features, not as validated optimum-range violations. This distinction is particularly important for cumin: the old reported heat-stress coefficient came from a different panel and is not a current confirmed result.

E. Baseline design and interpretation of R^2

The evaluation distinguishes a pooled performance measure from incremental explanation. A high pooled R^2 can arise when crop species and agricultural years have very different typical yields. A crop–year baseline estimates the mean log yield for the relevant crop and year using training data only. A complex model must beat this baseline on withheld districts to demonstrate added prediction from environmental covariates. The project also reports within-crop–year comparisons and an exploratory weather-anomaly analysis; those experiments did not establish a reliable incremental environmental yield signal.

For predicted values $\hat{y}_i$, the root mean squared error and mean absolute error are

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|. \tag{5}$$

For binary cultivation, AUC measures pairwise discrimination, AP summarizes precision–recall behavior and the Brier score measures mean squared probability error. All three are reported because ranking alone does not establish probability calibration.

TABLE IV
FIVE-BLOCK CULTIVATION OCCURRENCE VALIDATION

Model	AUC	AP	Brier
Crop only	0.7436	0.8663	0.1941
Crop + environment	**0.9063**	**0.9508**	**0.1149**

F. Hierarchical structure and posterior diagnostics

The Bayesian model partially pools crop-specific coefficients, allowing shared information while retaining crop heterogeneity. District intercepts account for persistent between-district differences and agricultural-year terms absorb broad temporal shifts. Heavy-tailed residuals reduce sensitivity to extreme yield values. A positive posterior coefficient is not evidence of a causal benefit unless confounding and model specification are addressed. We inspect $\widehat{R}$, effective sample sizes, divergent transitions and posterior intervals. The current archived summaries establish acceptable sampling behavior but do not provide a full posterior predictive calibration study.

G. Spatial validation and uncertainty

The V9 experiment assigns each district to exactly one of five provisional geographic blocks. All rows from a withheld block are predicted by models fitted on the other blocks. This is stricter than leave-one-district-out because a model cannot borrow information from a nearby held-out district within the same block. The reported bootstrap confidence intervals resample districts from fixed out-of-fold predictions; they quantify one form of sampling uncertainty but do not incorporate alternative block designs, source revisions or future-year shifts.

The spatial interpolation target is the observed district cultivation rate, averaged over the crop–year records represented for that district. The Kriging test fits a spherical variogram using only training-block districts. An inverse-distance weighting baseline uses great-circle distance with inverse-square weights. Both are compared on the identical 33 held-out district targets. Kriging variance is exported, but coverage calibration and blockwise error maps are not yet sufficient to call the interpolated surface a validated decision-support map.

IV. RESULTS

A. Cultivation occurrence

The leave-one-district-out (LODO) logistic classifier achieved AUC 0.9347, compared with 0.7396 for crop identity alone (Fig. 1). Under the more demanding five-block geographic test, the environmental-plus-crop model achieved AUC 0.9063, AP 0.9508, and Brier score 0.1149, versus crop-only AUC 0.7436, AP 0.8663, and Brier 0.1941. The observed AUC improvement was 0.1627 (district-bootstrap 95% interval: 0.0811–0.2396). These metrics quantify prediction of reported cultivation labels, not intrinsic land suitability.

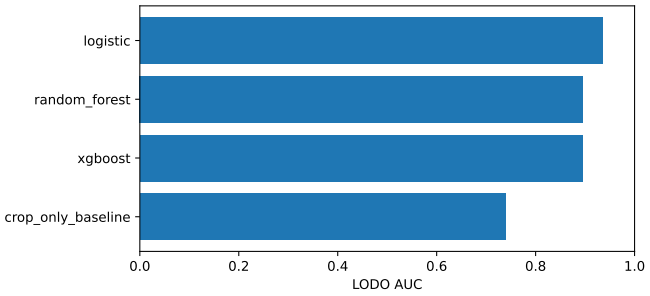


Fig. 1. Leave-one-district-out cultivation AUC by model. The more conservative five-block result is reported separately in the text.

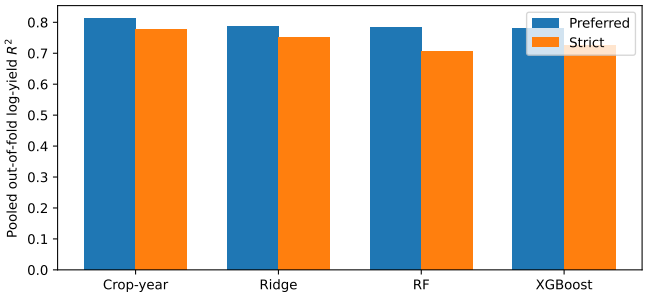


Fig. 2. Pooled out-of-fold log-yield R^2 for preferred and strict source selections. Crop–year baselines outperform all evaluated ML models.

B. Yield models and environmental associations

On the preferred yield subset, the crop–year baseline achieved pooled out-of-fold $R^2 = 0.8133$, exceeding ridge (0.7868), random forest (0.7827), and XGBoost (0.7815). The same ordering held on the strict subset: baseline 0.7781, ridge 0.7512, random forest 0.7062, and XGBoost 0.7249 (Fig. 2). Thus, no tested environmental ML model improved upon the baseline. A supplementary rainfall/humidity anomaly model also did not improve pooled yield prediction.

The preferred Bayesian model reported maximum $\widehat{R} = 1.00$, minimum bulk effective sample size 988, and zero divergences; the strict model reported 1.01, 718, and zero, respectively. The preferred shared slopes for standardized temperature, rainfall, and pH were -0.036 (94% highest density interval [HDI] $-0.096, 0.014$), -0.154 (HDI $-0.563, 0.193$), and -0.019 (HDI $-0.091, 0.060$). All three intervals overlap zero. Crop-specific slopes vary, and some are sensitive to source selection. These results do not demonstrate universal environmental effects on yield.

C. District zoning and geographic interpolation

Three environmental zones produced a silhouette score of 0.323. Their descriptive association with district cultivation rate was $\eta^2 = 0.526$ with exploratory permutation $p = 0.001$. Because zones and observations share geography, this test is not a substitute for independent spatial validation.

In five-block evaluation, ordinary Kriging achieved RMSE 0.2040 and mean absolute error (MAE) 0.1488, improving on IDW (RMSE 0.2551, MAE 0.2083) and the training-mean

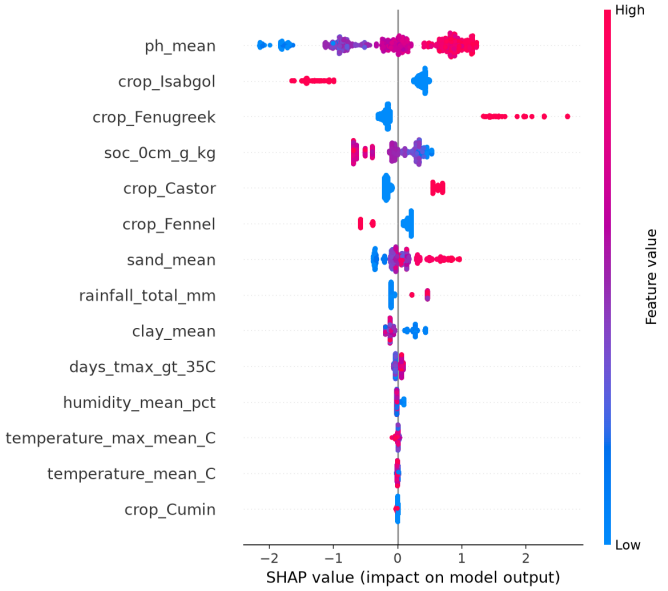


Fig. 3. Descriptive SHAP feature contributions from the existing ML experiment. These are model explanations, not causal effects or independently validated agronomic thresholds.

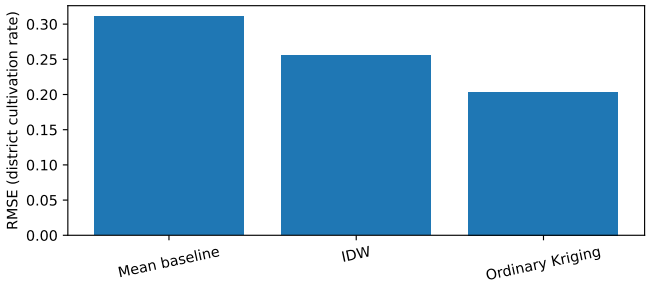


Fig. 4. Five-geographic-block interpolation errors for 33 district cultivation rates. Lower RMSE is better.

baseline (RMSE 0.3107, MAE 0.2618), as shown in Fig. 4. Kriging's RMSE was approximately 34.3% below the mean baseline. This is an encouraging district-scale interpolation result, not a validated field-scale hotspot map.

D. Reliability of the zero-label audit

The Stage 1 quality summary identifies 696 positive cultivation labels and 335 zero labels. The archived source-verification summary reports all 335 zeros matched their source APY cells, with no unresolved numerical mismatches. This result is stronger than merely checking that $A = 0$ implies $P = 0$ inside the model panel. However, it is not an independent audit of all 1,031 panel rows, and it cannot rule out underreporting or administrative differences in what qualifies as a cultivated crop. For transparency, the original source-file hashes, page or cell addresses, and review log should accompany a public data release.

E. Classification robustness across geographic splits

Under leave-one-district-out evaluation, logistic regression reached AUC 0.9347; random forest and XGBoost achieved

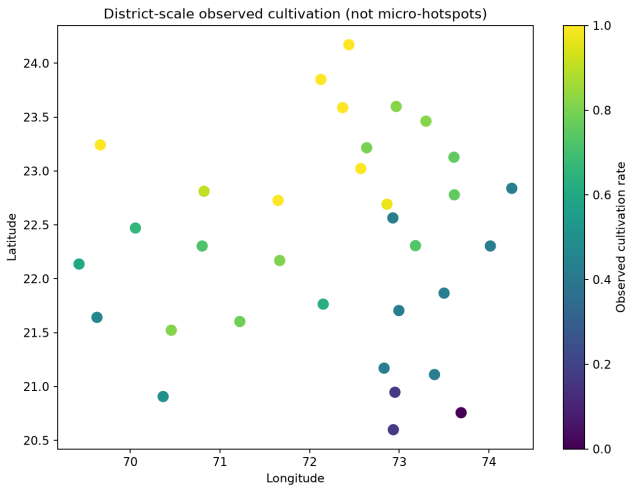


Fig. 5. Observed district-level cultivation rates at representative locations. Points do not depict within-district crop distributions.

0.8956 and 0.8951, respectively. The crop-only baseline reached 0.7396. The ranking shows that a comparatively simple linear decision surface generalized better than the evaluated tree ensembles in this setting. In the five-block test the logistic model's AUC declined to 0.9063, a reduction consistent with the more demanding geographic holdout. Its bootstrap AUC interval was 0.8604–0.9414, compared with 0.6824–0.8051 for crop identity alone. The interval for the difference was 0.0811–0.2396. These intervals are conditional on the fixed data and block scheme; no external-year test is reported.

F. Yield and sensitivity analyses

Table V shows that the crop–year mean is consistently the strongest evaluated log-yield predictor. This reverses the impression that a high pooled R^2 by itself establishes environmental explanatory power. In the V8 supplementary experiment, adding the tested rainfall and humidity anomalies reduced preferred-source pooled R^2 from 0.8133 to 0.8108 and strict-source R^2 from 0.7781 to 0.7750. The tested anomaly set was limited and is not a comprehensive test of all possible weather responses. A null incremental result should be reported rather than hidden.

G. Bayesian source sensitivity

The preferred Bayesian run used 682 observations and the strict run 540. Both had zero reported divergences and minimum bulk effective sample size above 700. The pooled coefficients in Table VI all have 94% HDIs crossing zero. In the strict model, shared slope means were -0.046, -0.319 and -0.026 for the same parameter indices, again with intervals spanning zero. Individual crop-specific terms can have narrower intervals and shift between source subsets, so any crop-level agronomic interpretation requires exact predictor mappings and a posterior predictive check. In particular, the earlier course report's strong cumin heat and castor pH interpretations arose from a different dataset and should not be carried forward as V6 findings.

TABLE V
OUT-OF-FOLD YIELD PREDICTION: ENVIRONMENTAL MODELS DO NOT SURPASS THE CROP–YEAR BASELINE

Source subset	Model	n	Pooled log-yield R^2	MAE (log yield)
Preferred	Crop–year baseline	682	**0.8133**	**0.1363**
Preferred	Ridge	682	0.7868	0.1512
Preferred	Random forest	682	0.7827	0.1522
Preferred	XGBoost	682	0.7815	0.1549
Strict	Crop–year baseline	540	**0.7781**	**0.1373**
Strict	Ridge	540	0.7512	0.1546
Strict	Random forest	540	0.7062	0.1671
Strict	XGBoost	540	0.7249	0.1627

TABLE VI
BAYESIAN POOLED COEFFICIENT SUMMARIES (PREFERRED SOURCE)

Parameter index	Posterior mean	94% HDI
Shared slope 0	-0.036	$[-0.096, 0.014]$
Shared slope 1	-0.154	$[-0.563, 0.193]$
Shared slope 2	-0.019	$[-0.091, 0.060]$

Parameter indices are retained from the archived PyMC summary; interpretation requires matching the exact feature ordering in the code.

H. Zoning and Kriging performance

Ward clustering formed three descriptive environmental groups. Its silhouette score was 0.323, indicating moderate rather than sharply separated structure. The observed association between group membership and cultivation rate was $\eta^2 = 0.526$ with an exploratory permutation $p = 0.001$. Because geographic structure can induce apparent separation, this is not a causal or operational zoning result.

Under five geographic blocks, the 33-district ordinary Kriging predictions achieved RMSE 0.203961 and MAE 0.148760. IDW reached RMSE 0.255122 and MAE 0.208272; the training-block mean yielded RMSE 0.310664 and MAE 0.261811. Thus Kriging reduced RMSE by about 34.3% versus the mean baseline and 20.1% versus IDW. This performance refers to interpolation of district cultivation *rates*, not yield, and does not imply that pixels between centroids are validated crop-suitability predictions.

V. DISCUSSION

The central finding is a divergence between two predictive tasks. Crop identity and environmental descriptors are informative for reported cultivation occurrence across withheld regions, whereas the same general data do not improve district yield prediction beyond crop–year averages. This does not imply that weather or soil is unimportant agronomically. Instead, district aggregates may mask irrigation, farm management, cultivar choice, soil heterogeneity, and spatially variable exposure. These candidate explanations were not independently tested here.

The cultivation result also deserves careful interpretation. Government-reported zeros have been checked against the source tables in the project audit, but official reporting is not a ground survey of every field. Weather is point-sampled by district, soil is model-based, the 33-district geographic policy has historical limitations, and some coordinate-review flags remain. The five spatial blocks are an internal stress test, not a new-state or future-year deployment evaluation. Spatial interpolation of only 33 rates is especially sensitive to block configuration and distance structure.

The Bayesian results reinforce this caution: favorable sampler diagnostics indicate that the fitted posterior was explored adequately, not that the specified predictors explain yield or validate traditional crop requirements. Likewise, descriptive SHAP attributions do not justify causal recommendations. More detailed agronomic stress functions would require separately verified crop-specific thresholds and growth-stage timing.

A. Why presence is easier to predict than yield

Cultivation presence is a coarse outcome: a crop is either reported or not reported in a district-year. It can reflect long-standing regional crop specialization, agroclimatic suitability, local institutions, irrigation investment and market networks. Yield is a finer outcome, conditional on cultivation, and depends on farm management, sowing dates, cultivar, water application, pest pressure and the distribution of actual cultivated plots. The present model measures only a fraction of these factors. Therefore the contrast between strong presence discrimination and weak incremental yield skill is plausible, but the mechanisms cannot be established with these observational data alone.

The findings should not be interpreted as "climate does not affect yield." A weather variable averaged over one point and one candidate growing window may miss short-duration stress during flowering, moisture availability from irrigation or spatial variability inside a district. A soil raster may accurately describe broad gradients without capturing field-level texture and drainage. Measurement error in both predictors and official yield targets can attenuate estimated relationships. These limitations motivate targeted data collection rather than arbitrary model complexity.

B. How the study differs from soil-only recommendation work

The Karnataka medicinal-plant literature motivates geographic plant matching, but this project uses historical government APY outcomes, explicit positive and zero-area labels, a crop-season weather panel, two government-source eligibility rules and withheld-region evaluation. This combination is a

more defensible methodological distinction than simply claiming that Gujarat has never been studied. We do not claim a novel machine-learning algorithm. The paper instead contributes an empirically evaluated distinction between predicting the observed cultivation footprint and predicting realized productivity. A comparison with an independent Gujarat or neighboring-state dataset would be required to assess wider generalization.

C. Implications for agricultural decision support

District-level cultivation prediction can be useful for hypothesis generation and regional monitoring. It cannot, on its own, recommend that an individual farmer grow a crop on a particular plot. A district with low reported cultivation may have suitable soils but unfavorable markets; a district with high cultivation may rely on irrigation or intensive management. The current zoning and Kriging outputs should therefore be treated as exploratory spatial summaries. Responsible deployment would require field surveys, land-use masks, irrigation data, uncertainty communication and prospective evaluation with local agronomists.

D. Threats to validity

Four threats deserve particular emphasis. *Target validity*: source-table zeros describe reported absence, not independently surveyed crop absence. *Spatial dependence*: some districts share the same climate grid value, so the nominal 33 districts do not supply 33 independent weather environments. *Historical comparability*: present-day polygons may not correspond exactly to historical administrative reporting units. *Outcome confounding*: management, irrigation and markets are omitted. The project also includes source conflicts, coordinate-review flags and modeled rather than measured soil properties. These factors constrain causal and agronomic interpretation even when cross-validation scores are favorable.

VI. Reproducibility, Limitations, and Future Work

The underlying project contains source-linked APY evidence, two yield-selection rules, seasonal NASA POWER features, gridded soil attributes, code for four stages, and V8–V9 evaluation outputs. The present figures and tables are based on those archived results. A final publication should additionally release source-to-cell verification logs for nonzero yields, complete data dictionaries and source licenses, fixed geographic block definitions, and reproducible environment specifications.

Priorities for follow-up are district-year irrigation and management data, alternate gridded climate sources, posterior predictive checking, uncertainty calibration for Kriging, and external temporal or regional validation. More models alone are unlikely to resolve missing explanatory variables. Any eventual recommendation system should be tested against independent agronomic evidence before operational use.

A. Reproducible artifact structure

The executable project separates four primary data tables, stage scripts, model outputs and later audits. The Stage 1 JSON records panel sizes and coordinate-review counts. Stage 2 exports posterior CSV summaries, diagnostics and NetCDF posterior objects. Stage 3 exports out-of-fold predictions for presence and yield models and a descriptive SHAP figure. Stage 4 exports zones, interpolation predictions and a map. V8 evaluates anomaly features and additional diagnostics; V9 exports geographic-block predictions and conditional bootstrap intervals. The final source audit and spatial-block Kriging run are separate from the original four stages. This provenance separation is essential for understanding which results were obtained from which release.

B. Publication checklist and future extensions

Before journal submission, the authors should reconcile the remaining coordinate flags, inspect posterior predictive distributions, quantify uncertainty calibration of Kriging, verify a representative sample of positive-area and yield records against original government tables, and document source licenses. A robust future experiment would combine district-year irrigation shares, alternative climate grids such as ERA5-Land or CHIRPS, spatially explicit cultivated-area masks, and independently observed agronomic calendars. A temporal holdout on later years or external evaluation outside Gujarat would be more persuasive than repeated optimization on the same 33 districts. Crop-specific stress indices should only be tested after their thresholds and units are independently sourced. These are prospective research steps, not experiments claimed to have been completed.

VII. Conclusion

Across five Gujarat crops, environmental and crop features predict *reported cultivation occurrence* substantially better than crop identity alone under geographic-block testing. The tested environmental yield models do not outperform crop–year averages, and pooled Bayesian environmental effects remain uncertain. District-scale Kriging improves interpolation error relative to mean and IDW baselines, but does not establish farm-level hotspots. Together these findings motivate a distinction between predicting agricultural distribution and explaining productivity. The study provides a reproducible basis for further validation rather than a claim of publication-certified agronomic suitability.

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